

Equilibrium Concepts

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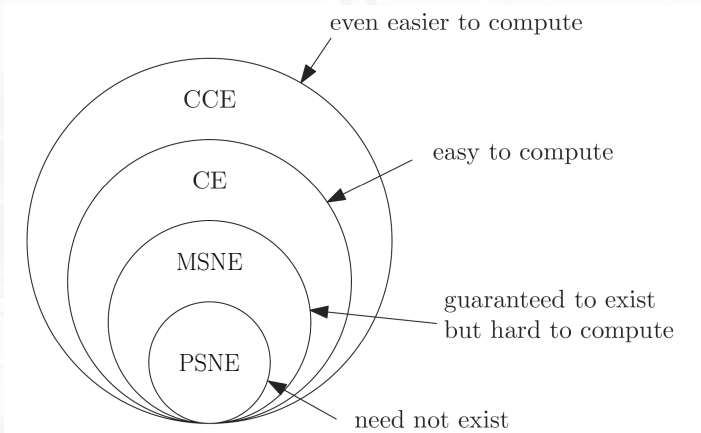


Outline

- 1 Cost Minimization and Payoff Maximization
- 2 Pure Strategy Nash Equilibria (PSNE)
- 3 Mixed Nash Equilibria (MSNE)
- 4 Correlated Equilibria (CE)
- 5 Coarse Correlated Equilibria (CCE)



A hierarchy of equilibrium concepts



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Cost-Minimization Games

A cost-minimization game has the following ingredients:

- a finite number of k agents;
- a finite set S_i of pure strategies for each agent i ;
- a nonnegative cost function $C_i(\mathbf{s})$ for each agent i .
 - $\mathbf{s} \in S_1 \times S_2 \times \cdots \times S_k$: a **state**, **strategy profile** or **outcome**.

For example, the facility location game, congestion (routing) game, etc.



Payoff-Maximization Games

A **payoff-maximization** game has the following ingredients:

- a finite number of k agents;
- a finite set S_i of pure strategies for each agent i ;
- a nonnegative **payoff** function $\pi_i(\mathbf{s})$ for each agent i .
 - $\mathbf{s} \in S_1 \times S_2 \times \cdots \times S_k$: a **state**, **strategy profile** or **outcome**.

For example, the Rock-Paper-Scissors game, two-party election game, etc.



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Pure (Strategy) Nash Equilibrium (PSNE)

Pure Nash Equilibrium (PSNE)

A strategy profile $\mathbf{s}$ of a cost-minimization game is a pure Nash equilibrium (PSNE) if for every agent $i \in \{1, 2, \dots, k\}$ and every unilateral deviation $s'_i \in S_i$,

$$C_i(\mathbf{s}) \leq C_i(s'_i, \mathbf{s}_{-i}).$$

- $\mathbf{s}_{-i}$: the vector $\mathbf{s}$ with the i th component removed.



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Mixed Nash Equilibrium (MSNE)

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Distributions $\sigma_1, \dots, \sigma_k$, over strategy sets $S_1, \dots, S_k$ respectively, of a cost-minimization game constitute a mixed Nash equilibrium (MSNE) if for every agent $i \in \{1, 2, \dots, k\}$ and every unilateral deviation $s'_i \in S_i$,

$$\mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(\mathbf{s})] \leq \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(s'_i, \mathbf{s}_{-i})].$$

- σ : the product distribution $\sigma_1 \times \sigma_2 \times \dots \times \sigma_k$.

Product of Mixed Strategies

Player 2

		q_1 rock	q_2 scissors	q_3 paper	→ probabilities
Player 1	p_1 rock	$p_1 q_1$ 0, 0	$p_1 q_2$ 1, -1	$p_1 q_3$ -1, 1	
	p_2 scissors	$p_2 q_1$ -1, 1	$p_2 q_2$ 0, 0	$p_2 q_3$ 1, -1	
	p_3 paper	$p_3 q_1$ 1, -1	$p_3 q_2$ -1, 1	$p_3 q_3$ 0, 0	

↓ probabilities

$$\begin{cases} p_1 + p_2 + p_3 = 1. \\ q_1 + q_2 + q_3 = 1. \end{cases}$$

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Correlated Equilibrium (CE)

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A **distribution σ on the set $S_1 \times S_2 \times \dots \times S_k$ of outcomes** of a cost-minimization game is a correlated equilibrium (CE) if for every agent $i \in \{1, 2, \dots, k\}$ and every unilateral deviation $s'_i \in S_i$,

$$\mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(\mathbf{s}) \mid \mathbf{s}_i] \leq \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(s'_i, \sim_{-i}) \mid \mathbf{s}_i].$$



Stop or Go?

Matrix of costs

	Stop	Go
Stop	1, 1	1, 0
Go	0, 1	5, 5

- Two PSNE.

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	Stop	Go
Stop	prob. = 0 1, 1	prob. = 1/2 1, 0
Go	prob. = 1/2 0, 1	prob. = 0 5, 5

- A CE for example.
- Cannot correspond to a MSNE.



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Game of Chicken

- A.k.a. Hawk-Dove Game.
 - A model of conflict for two players.

	Dare	Chicken
Dare	0, 0	7, 2
Chicken	2, 7	6, 6

- Two PSNE & One MSNE.
- The expected utility of each player in the MSNE:

$$\frac{1}{3} \cdot \frac{2}{3} \cdot 7 + \frac{2}{3} \cdot \frac{1}{3} \cdot 2 + \frac{2}{3} \cdot \frac{2}{3} \cdot 6 = \frac{14}{3}.$$



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Game of Chicken

- A correlated equilibrium.
- Check that it is an equilibrium if a player is assigned “Dare”.
- Check that it is an equilibrium if a player is assigned “Chicken Out”.

	Dare	Chicken
Dare	prob. = 0 0, 0	prob. = 1/3 7, 2
Chicken	prob. = 1/3 2, 7	prob. = 1/3 6, 6

- The expected utility for each player:
 $7 \cdot (1/3) + 2 \cdot (1/3) + 6 \cdot (1/3) = 5.$



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$$\mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(\mathbf{s})] \leq \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(s'_i, \mathbf{s}_{-i})].$$

CE $\subseteq$ CCE?

$$\begin{aligned} \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(\mathbf{s})] &= \sum_{a \in S_i} \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(\mathbf{s}) \mid s_i = a] \Pr[s_i = a] \\ &\leq \sum_{a \in S_i} \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(s'_i, \mathbf{s}) \mid s_i = a] \Pr[s_i = a] \\ &= \mathbb{E}_{\mathbf{s} \sim \sigma}[C_i(s'_i, \mathbf{s}_{-i})] \end{aligned}$$



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CCE Example

	A	B	C
A	prob. = 1/3 1, 1	-1, -1	0, 0
B	-1, -1	prob. = 1/3 1, 1	0, 0
C	0, 0	0, 0	prob. = 1/3 -1.1, -1.1

- The payoff for each player (playing according to this distribution):
 $\frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 1 - \frac{1}{3} \cdot 1.1 = 0.3.$
- A player playing fixed A or B while the opponent randomized according to this distribution: $\frac{1}{3} \cdot 1 - \frac{1}{3} \cdot 1 + \frac{1}{3} \cdot 0 = 0.$
- A player playing fixed C while the opponent randomized according to this distribution: $\frac{1}{3} \cdot 0 + \frac{1}{3} \cdot 0 + \frac{1}{3} \cdot (-1.1) < 0.$



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CCE Example

	A	B	C
A	$\text{prob.} = 1/3$ $1, 1$	$-1, -1$	$0, 0$
B	$-1, -1$	$\text{prob.} = 1/3$ $1, 1$	$0, 0$
C	$0, 0$	$0, 0$	$\text{prob.} = 1/3$ $-1.1, -1.1$

- A player playing fixed C and the strategy profile follows this distribution:
 $\Rightarrow$ deviation is possible.
 - Not a CE.

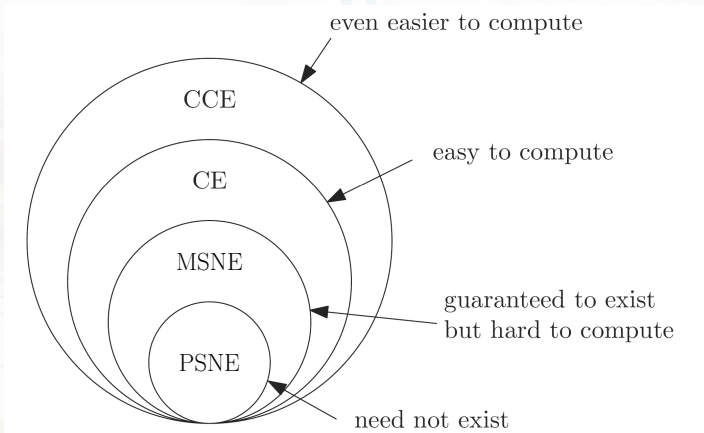
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Discussions

